

Harpreet Singh

Computer Engineering Student
Embedded Systems | FPGA | Systems Software

hs3732065@gmail.com | +1 (778) 998-1236

Vancouver, BC, Canada

linkedin.com/in/harpreet-singh-4523661a0 | github.com/harpreeto6

EDUCATION

University of British Columbia (UBC)
BAsC, Computer Engineering (Vancouver, BC)

Sept 2024 to Dec 2027 (expected)

EXPERIENCE

Software Development Engineer Intern, Amazon

May 2026 to Jul 2026

- Developing production software with emphasis on implementation, debugging, testing, maintainability, and code review in a large-scale codebase.
- Strengthening systems-level engineering habits by reasoning about reliability, integration boundaries, and existing architecture.

MOps Software Developer, UBC ORBIT Satellite Design Team

Jan 2026 to Present

- Building mission-operations software across frontend, backend, deployment, and testing workstreams for a satellite engineering team.
- Collaborating with other ORBIT subteams through Git/GitHub, issue tracking, code review, and engineering-focused development practices.

PROJECTS

RISC-V Processor with Memory-Mapped I/O

Academic Project

SystemVerilog | RISC-V | FPGA I/O | Memory-Mapped Registers | DE10-Lite

- Implemented a single-cycle RISC-V-style processor top level with instruction/data memories, controller/datapath wiring, branching, loads/stores, and ALU control.
- Added memory-mapped I/O for DE10-Lite switches, LEDs, and seven-segment displays using address-decoded write/read behavior.

FPGA Digital Safe System

Academic Project

SystemVerilog | FSMs | Quartus | Questa | DE10-Lite | Seven-Segment Displays

- Designed a digital safe with open/locked states, password save, attempt capture, reset behavior, match detection, and hint LEDs.
- Built finite-state control logic and display encoding to show safe status across six seven-segment displays.

Keypad Scanner and Debounced Input Interface

Academic Project

SystemVerilog | Synchronizers | Debouncing | Active-Low I/O | Key Decode

- Implemented active-low row scanning, column sampling, key-code decoding, key-held state tracking, and valid-key signaling.
- Added synchronizer and debounce logic to stabilize asynchronous keypad input before driving downstream FPGA logic.

Low-Level Systems and Networking Labs

Academic Coursework

C/C++ Concepts | Concurrency | Virtual Memory | Processes | Networking | Wireshark

- Studied systems topics including malloc, linking, files, processes, concurrency, garbage collection, and virtual memory through CPEN 212 materials.
- Completed networking lab work involving protocol behavior and packet-level analysis for DHCP, DNS, NAT, and TCP congestion-control experiments.
- Practiced packet-level debugging and protocol reasoning with Wireshark traces, plots, and lab evidence.

Sensum: Full-Stack Productivity Platform

Personal Project

Java 17 | Spring Boot | PostgreSQL | Next.js | TypeScript | Chrome MV3

- Built a backend-backed application with REST APIs, PostgreSQL persistence, JWT authentication, browser-extension telemetry, and user workflow state.

SKILLS

Languages: C++, C, Java, Python, SystemVerilog, TypeScript

Tools: Quartus, Questa, Git/GitHub, Docker, VS Code

Systems: FPGA, FSMs, RISC-V, Memory-Mapped I/O, Debugging

Software: Spring Boot, REST APIs, PostgreSQL, React / Next.js